CHAPTER 3

TRAFFIC SOURCE MODELS

3.1 Introduction

The Information Flow model captures the average traffic generated by a source, but it overlooks the finer temporal dynamics. While this model is useful for rough network resource dimensioning or assessing system stability, it falls short when analyzing more specific performance metrics such as delay, traffic peaks, or for accurately sizing certain network components, such as buffers. For these purposes, it is necessary to describe the offered traffic profile, i.e., the pattern with which data are actually generated by the source, through suitable *traffic models*.

According to the final design goal, a traffic model can actually capture different aspects of the traffic generated by a source. For example, to have an idea of the order of magnitude of the bitrate required from a trunk connection, it suffices to model just the long-term average traffic offered by all the data sources multiplexed over that trunk. Instead, to check whether the average transmission delay of a certain Information Flow is below a given threshold, the model also needs to account for the time dynamics of the traffic source. In both cases, the actual content of the traffic is irrelevant for the network design and dimensioning, and hence it is usually not considered in traffic modeling. If the purpose is to design access schemes to support traffic prioritization, instead, the source model should also differentiate the packets generated by the source according to their priority class.

In the rest of this chapter, we will first illustrate a general method to describe a traffic source by means of mathematical models. Successively, we will discuss some basic traffic source models, which can be used to check the performance of some network manage-

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ment protocols (MAC and scheduling algorithms, in particular), or to provide a first, rough dimensioning of the network resources. Finally, we will present some methodologies to verify the model validity.

3.2 Abstractions of an Information Flow

An Information Flow is typically associated to a logical *session* between two (or more) End Nodes that need to exchange information. For example, when a web browser contacts a web server to get a certain World-Wide Web (WWW) content, the two End Nodes establish an Hypertext Transfer Protocol (HTTP) session. During the session, the nodes will actively exchange data in certain time intervals, and remain idle in others (e.g., when the user consults the content of the downloaded webpage). The session ends when the client connects to a different web server or it is closed by the user. A different example of a logical session between two End Nodes may be that of a mailer fetching new emails from a mail server. In this case, the session starts with the request from the client to the mail server, and ends when all the emails are downloaded to the mailer application. Therefore, traffic exchange during a session may be irregular, with alternating periods of active data exchange with periods of inactivity. During the active periods, furthermore, the traffic exchanged by the End Node can be constant or variable, according to the type of application.

Depending on the purpose of the traffic model, therefore, we can describe an Information Flow at different levels of abstractions, as depicted in Fig. 3.1, i.e.,

- *Session level*: ithe model only describes when an Information Flow starts and ends, and what is the average generated traffic. In other words, the model describes a logical *session* between two End Nodes;
- *Activity level*: the model describes the active and idle periods during a session, by defining the duration of such periods (usually, in statistical terms) and the average traffic generated in each period;
- *Packet level*: the model describes in details the packet generation pattern and packet size distribution during active and idle periods.

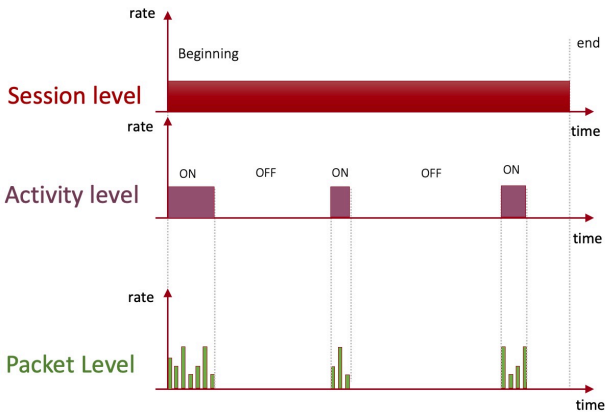


Figure 3.1 Graphical representation of the different levels of abstraction of a source model.

In the following, we first introduce a very general model that can be used to represent a large variety of traffic sources. Then, we will present some very common traffic source models that are often used in simulation or for mathematical analysis of communication systems.

3.3 A general model: Marked and Weighted Point Process

A Point Process (PP) is a stochastic model used to represent the occurrence of discrete random events in some continuous domain, typically the time axis or a Cartesian plane. In general, events (and their continuous domains) can be of the most diverse nature.

In the context of source modeling, the domain is typically the time axis, and the events are connected to the generation of new traffic. Therefore, depending on the level of abstraction of the source traffic model, events can correspond to the beginning or ending of a session, or to the beginning/ending of active/inactive periods, or to the generation of new packets. In any case, we will refer to any such events as an *arrival*. Furthermore, without loss of generality, we can describe the arrival process from time zero onward.

Mathematically, the PP can then be represented by the set $\mathcal{T} = \{\tau_i\}$ of arrival instants, where $\tau_i \in \mathcal{R}^+$ indicates the time at which the ith arrival occurs, with $i = 1, 2, \ldots$. An equivalent representation of the process can be given in terms of the so-called *inter-arrival times*, i.e., the set $\mathcal{A} = \{A_i\}$, where $A_i = \tau_i - \tau_{i-1}$ is the time between the $i-1$st and the ith arrival. A third, equivalent way to represent the PP is by means of the associated *counting process* $N(t) : \mathcal{R} \to \mathcal{N}$, which represents the number of arrivals of the PP in the time interval $(0, t]$, i.e.,

$$N(t) = \arg\max_n \{\tau_n \leq t\} = \sum_{k=0}^{\infty} \chi\{\tau_k \leq t\} \ . \tag{3.1}$$

For ease of notation, the number of arrivals occurring in a given time interval $(t_1, t_2]$, with $0 \leq t_1 \leq t_2$, will be indicated as $N(t_1, t_2) = N(t_2) - N(t_1)$.

A Point Process (PP) can be extended to include additional information associated with each event, resulting in a *marked point process* or a *weighted point process*, as explained next.

3.3.1 Marked Point Process

A generalization of the PP is the Marked Point Process, where each arrival event can be associated to a certain numerousness, indicated by a compound process $\mathcal{B} = \{B_i\}$, $i = 0, 1, 2, \ldots$, where $B_i \in \mathbb{N}_0$ is the *mark* associated to the ith arrival. Mathematically, the process can be represented as:

$$(\mathcal{T}, \mathcal{B}) \, , \quad \text{or equivalently} \quad (\mathcal{A}, \mathcal{B}) \, ,$$

where B_i is the mark associated to the ith arrival. The marked point process is called *independent* if the marks B_i are independent of the arrival times τ_i. Otherwise, it is said to be *dependent*.

The marks B_i can represent characteristics such as number of Information Units (IUs) generated at the ith arrival event, the type of the IUs, or the priority. In the following, we assume that the mark actually represents the *batch size*, i.e., the number of IU generated at each arrival event.

Consider, for example, an application that generates messages to be delivered to a certain destination. Such messages can consist of multiple packets that are passed to the network layer for transmission. Then, from the perspective of the network layer, the generation of a new message by the source can be modelled as an arrival that, however, brings along a bunch of packets to be transmitted, the numerousness of which is modelled by the process $\mathcal{B}$.

3.3.2 Marked and Weighted Point Process

A Marked and Weighted Point Process (MWPP) enriches the marked point process $(\mathcal{T}, \mathcal{B})$ with the *weight* process $\mathcal{W} = \{W_{i,k}\}$, which assigns a weight $W_{i,k}$ to each IU $k \in \{1, \ldots, B_i\}$ in the batch associated with arrival τ_i. The weight represents the significance or contribution of the IU for the system. For example, considering the kth IU in the ith arrival batch, $W_{i,k}$ can be the length in bits of the IU, or the energy consumption required for its transmission, or the computation required for its processing. Therefore, these weights are hence associated to a certain time of *workload*.

Therefore, a MWPP is a generalized representation of a PP that includes both marks and weights, and is hence defined by the triplet:

$$(\mathcal{T}, \mathcal{B}, \mathcal{W}) \quad \text{or equivalently} \quad (\mathcal{A}, \mathcal{B}, \mathcal{W})\,.$$

In this books, whenever the MWPP is used to model a packet generation process, the process $\mathcal{T}$ models the generation events, $\mathcal{B}$ indicates the batch sizes, i.e., the number of packets simultaneously generated at each arrival event,[1] and $\mathcal{W}$ represents the workload associated with each packet. In this case, the rate of the traffic generated by the application in a certain time interval $(t_1, t_2]$ can be expressed as

$$\mathrm{G}(t_1, t_2) = \frac{\sum_{i=N(t_1)+1}^{N(t_2)} \sum_{k=1}^{B_k} W_{i,k}}{t_2 - t_1}\,. \tag{3.2}$$

The MWPP offers a very general model that can be specialized in many different ways to provide a variety of specific traffic source models, some of which are described in the next sections.

Example 3.3 A Consider an application that generates messages of M bytes every T seconds. This source can be represented by a MWPP with parameters $\mathcal{A} = \{T\}$, $\mathcal{B} = \{1\}$, $\mathcal{W} = \{M\}$, since arrivals occur regularly with period T, at each arrival corresponds one single message, and the message size is M.

Suppose now that each message needs to be split in K packets to be forwarded across a certain data link layer. Then, from the perspective of the data link layer, the packet generation model is a MWPP with $\mathcal{A} = \{T\}$, $\mathcal{B} = \{K\}$, $\mathcal{W} = \{M/K\}$.

3.3.3 Renewal reward processes

A specific case of a PP is the Renewal Process, which requires the inter-arrival times A_i to be i.i.d. r.v.s with a common statistical distribution $p_{\mathcal{A}}(\cdot)$ (see Sec. **??**). Each arrival

[1] Typically, $B_i \geq 1$. However, it is sometimes convenient to consider $B_i \geq 0$, with $B_i = 0$ indicating events that do not bring any workload to the system. This "silent" event may correspond, e.g., to the generation of a packet that is dropped before transmission.

represents a *renewal time* of the process, as the statistical evolution of the process no longer depends on the past, but only on the nominal distribution of the inter-arrival times.[2]

When the distribution of the inter-arrival times is exponential with parameter λ, then the Renewal process is a Poisson Point Process (PPP) with parameter λ (see Sec. **??**). Accordingly, the number of arrivals in a time interval $(t_1, t_2]$ is a r.v. with Poisson distribution of parameters $\lambda(t_2 - t_1)$, and the arrivals in disjoint time intervals are independent.

Renewal-reward processes (see Sec. **??**) extend Renewal processes by associating *rewards* $\{R_n\}$ with events. In the context of traffic modeling, the reward can be bound to the batch size, or the workload, or both. The *elementary renewal theorem* provides a foundational result for analyzing the long-term behavior of such processes. If rewards are i.i.d. with expected value $\mathrm{E}[R]$, then the expected reward rate is given by:

$$\lim_{t\to\infty} \frac{1}{t} \sum_{n=1}^{N(t)} R_n = \frac{\mathrm{E}[R]}{\mathrm{E}[A]} .$$

This theorem links the long-term average reward to the expected reward per renewal and the expected renewal time.

Example 3.3 B Consider again the Example 3.3 B. Assuming $\mathcal{B}$ and $\mathcal{W}$ are independent of $\mathcal{A}$, the average rate of the traffic generated by the source can be expressed as the ratio between the average workload generated in a renewal interval and the mean duration of a renewal interval, i.e.,

$$\mathrm{G} = \frac{\mathrm{E}\left[\sum_{k=1}^{B_i} W_{k,i}\right]}{\mathrm{E}[A_j]} = \frac{\mathrm{E}[B_\ell]\,\mathrm{E}[W_{k,i}]}{\mathrm{E}[A_j]} . \tag{3.3}$$

3.4 Classification of Source Traffic Models

The behavior of traffic generated by sources in communication networks can be classified into several distinct models, each reflecting specific characteristics of data generation and transmission. These models are critical for understanding and analyzing network performance, as they help capture the diversity of traffic patterns exhibited by different applications. In this section we list the main models, which represent a first high-level classification of traffic generation patterns. This classification provides a foundation for selecting appropriate models based on the nature of the application or network scenario under consideration.

It is important to remark that the classification of a traffic source depends on the behavior of the traffic rate that, in turn, is defined with respect to a certain time interval. By changing the reference time interval, the classification of the traffic source can also change. For example, the traffic generated by a node that sends packets of slightly variable size at regular time intervals can be seen as continuous byte stream of constant rate if observed at time intervals much larger than the inter-packet time, while at a closer look, the traffic pattern appears more variable and dynamic.

[2]It should be noted that this condition is generally not satisfied at time instants other than the renewal times, unless the inter-arrival times are exponentially distributed.

3.4.1 Saturated sources

The saturated source is a theoretical model that is commonly used to test the performance of a system under heavy traffic conditions. Indeed, a *saturated source* assumes the source is always *backlogged*, i.e., with new packets ready for transmission as soon as permitted by the resource manager.

Mathematically, the model can be described as a MWPP with very small inter-arrival times $\{A_i \simeq 0\}$, unit marks $\{B_i = 1\}$, and constant or variable workload. Since in simulations it is not possible to generate zero inter-arrival times (which would require the generation of an infinite number of simultaneous arrivals), the saturated traffic source is typically simulated by generating a new packet as soon as the previous packet leaves the network interface, so as to always have a pending packet ready for transmission.

3.4.2 Constant Bit Rate

The CBR model is used to represent sources that generate constant-size packets periodically. Mathematically, a CBR source can be represented by a MWPP with $\mathcal{A} = \{A\}$, $\mathcal{B} = \{B\}$, $\mathcal{W} = \{W\}$, where A is the (constant) inter-arrival time, B is the (constant) number of packets generated at each arrival event, and W is the (constant) packet size.

This is the case, for example, of the traffic generated by a digital voice call coded with the G.711 coding scheme. In this case, the voice signal is sampled at 8 kHz, and each sample is quantized in 256 levels and represented with 8 bits, so that the output traffic from the voice coder is CBR with an average rate of 64 kbit/s. To be carried across the Internet, the bitstream is split into packets, whose (constant) size dependeds on the packetization interval. For G.711, the most common intervals are 10 ms, 20 ms, or 30 ms, corresponding respectively to packets of 80 bytes, 160 bytes, or 240 bytes. To be noticed that the actual packets at network layer are longer because of the control information (headers) added by the other protocols, namely the Real Time Protocol (12 bytes of header), UDP (8 bytes of header) and IP (20 bytes of header).

3.4.3 Variable Bit Rate

VBR models are widely used in case of multimedia sources, where the data rate is inherently tied to the characteristics of the content being transmitted. For example, in video streaming, the datarate required to transmit a sequence of frames encoded with some advanced video-compression technique can vary substantially from scene to scene, depending on the level of motion, the complexity of the scene, and the compression algorithm used. A static scene with minimal motion can be typically represented with fewer bits, while a fast-moving or highly detailed scene demands a significantly higher data rate. Such variability is efficiently captured using VBR models, which can also reflect real-world constraints such as bandwidth limitations or processing power at the source.

Beyond multimedia applications, VBR sources are also relevant in representing traffic from interactive or event-driven systems, such as online gaming, video conferencing, or Internet of Things (IoT) networks. In these contexts, the traffic generated often depends on external stimuli, such as user actions, environmental changes, or network feedback. For instance, an IoT device monitoring environmental conditions may transmit small amounts of data at regular intervals but produce larger bursts of traffic in response to detected anomalies. Similarly, in a video conferencing scenario, the traffic rate may increase during peri-

ods of active speech or high motion in the video feed and decrease during silence or low activity.

The mathematical modeling of VBR sources typically involves stochastic processes that can capture their dynamic behavior. The MWPP modeling of a VBR source requires at least one of the processes $\mathcal{A}$, $\mathcal{B}$ and $\mathcal{W}$ to be variable. Techniques such as Markov-Modulated Poisson Process (MMPP) (see later) are commonly employed to represent the variability in arrival rates and packet sizes. Additionally, advanced techniques like fractional Brownian motion and heavy-tailed distributions can be applied to model the self-similar and long-range dependent properties often observed in VBR traffic.

3.4.4 Bursty Sources

Bursty sources are a class of traffic models that exhibit alternating periods of high and low activity, often characterized by abrupt changes in transmission rates. This behavior, referred to as *burstiness*, is a fundamental property observed in many real-world applications. Bursty traffic patterns are inherently irregular and unpredictable, making them challenging to model and analyze, yet they play a crucial role in understanding and designing communication networks.

The bursty nature of these sources stems from the interaction between user behavior, application requirements, and the underlying protocols. For example, in web browsing, users typically alternate between downloading content, such as HTML pages, images, or videos, and periods of inactivity while consuming the downloaded material. During active periods, large volumes of data may be transmitted in quick succession, followed by intervals of little or no traffic. Similar patterns are observed in other applications, such as online gaming, where bursts of data are generated in response to player actions, or file transfer protocols, which may transmit large files in segmented bursts.

Quantifying burstiness is essential for accurately characterizing traffic patterns and assessing their impact on network performance. Several metrics are used to describe the degree of burstiness in a traffic source. One common measure is the *peak-to-mean ratio*, which compares the maximum transmission rate observed during a burst to the average rate over time. A higher ratio indicates more pronounced bursts.

Another important metric is the *coefficient of variation* (CV) of interarrival times, defined as the ratio of the standard deviation to the mean. A CV greater than one typically indicates significant variability in arrival times, which is a hallmark of bursty traffic.

Additionally, the *index of dispersion* (also known as the Fano factor) measures the variance-to-mean ratio of the number of arrivals in a given time interval, providing further insight into the temporal variability of traffic.

Mathematical models of bursty sources often employ stochastic processes to represent alternating periods of high and low activity, as explained in the following sections.

3.4.5 On-Off model

The On-Off model is one of the simplest representations of bursty traffic. In this model, the source alternates between two states: an "ON" state, during which CBR or VBR traffic is actively generated, and an "OFF" state, during which no traffic is produced.

The On-Off source can be modelled as a stochastic process with two states, namely ON and OFF. The time spent in a state, before moving to the other, is named *sojourn time*, and is typically assumed to be random. Then, let t_{ON} and t_{OFF} be the r.v.s representing the sojourn times in the ON and OFF states, respectively. Typically, the sojourn times

are assumed to be i.i.d. for each state, and usually independent between states. We indicate their Cumulative Distribution Functions (CDFs) as $F_{ON}(t)$ and $F_{OFF}(t)$, with mean $T_{\text{ON}} = \mathrm{E}[t_{\text{ON}}]$ and $T_{\text{OFF}} = \mathrm{E}[t_{\text{OFF}}]$, respectively. Common probability distributions for these r.v.s are exponential, Pareto, or heavy-tailed. Extensions of the On-Off process, such as the MMPP (see later), allow for more complex and realistic representations of bursty traffic.

We observe that an On-Off cycle represents a renewal interval of duration $t_{\text{Cycle}} = t_{\text{ON}} + t_{\text{OFF}}$ and mean $T_{\text{Cycle}} = T_{\text{ON}} + T_{\text{OFF}}$. If traffic is generated only during the ON period, we can compute the peak rate of the source (that is, the traffic generation rate during the active phase) by analizing the mean traffic of a modified process, simply obtained by removing the OFF periods. The resulting modified process will therefore always be in the active phase. Therefore, if the packet generation process is described by a MWPP $(\mathcal{A}, \mathcal{B}, \mathcal{W})$, then the peak rate (i.e., , the traffic rate in the ON phase) is given by

$$\mathrm{G}_{\text{ON}} = \frac{\mathrm{E}[B_\ell]\,\mathrm{E}[W_{i,k}]}{\mathrm{E}[A_j]}\,. \tag{3.4}$$

The overall average traffic rate generated by the On-Off source, instead, can be computed by scaling (3.4) by the probability that the source is in the ON state. Since each On-Off cycle represents a renewal event, we can model the number of bits generated in a renewal interval as the reward gained during that renewal cycle and then apply the renewal reward theorem. This gives:

$$\mathrm{G} = \frac{\mathrm{E}\left[\sum_{i=1}^{N(t_{\text{ON}})} \sum_{k=1}^{B_i} W_{k,i}\right]}{\mathrm{E}[t_{\text{Cycle}}]} = \frac{\frac{T_{\text{ON}}}{\mathrm{E}[A_j]}\mathrm{E}[B_\ell]\,\mathrm{E}[W_{i,k}]}{T_{\text{ON}} + T_{\text{OFF}}} = \frac{\mathrm{G}_{\text{ON}} T_{\text{ON}}}{T_{\text{ON}} + T_{\text{OFF}}}\,. \tag{3.5}$$

A straightforward extension of the On-Off model assumes that some traffic can also be generated during the Off state, at an average rate $\mathrm{G}_{\text{OFF}} < \mathrm{G}_{\text{ON}}$. In this case, the average traffic rate generated by the source is given by

$$\mathrm{G} = \frac{\mathrm{G}_{\text{ON}} T_{\text{ON}} + \mathrm{G}_{\text{OFF}} T_{\text{OFF}}}{T_{\text{ON}} + T_{\text{OFF}}}\,, \tag{3.6}$$

as can be readily derived using the same reasoning as above.

3.5 Markovian Traffic Source Models

Markovian traffic source models are a class of stochastic models widely used to represent traffic behavior in communication networks. These models rely on the mathematical framework of Markov processes, in which the future state of the system depends only on its current state. This property makes them particularly suitable for describing traffic whose evolution can be represented through a finite or countable set of operational states. Markovian models can capture a variety of traffic characteristics, including variability, burstiness, and transitions between different activity states.

In general terms, a Markovian traffic source model can be described by a Markov chain, that is, a discrete-time Markov process ω_n taking values in a discrete state space $\Omega \subseteq \mathbb{Z}$. Each state is associated with a different operational mode of the traffic source (e.g., idle, active, or bursty), corresponding to a particular traffic-generation pattern. Transitions between states represent changes in the operational mode of the source, such as a transition from an idle state to an active state.

In a Markov model, state transitions are stochastic, and their probabilities are described by the transition probability matrix (or simply transition matrix) $\mathbf{P} = [P_{i,j}]$, where $P_{i,j} = \mathrm{P}\left[\omega_{n+1} = j | \omega_n = i\right]$ is the probability that the source moves from state i to state j in one time step. If the transition matrix $\mathbf{P}$ does not depend on the time step n, the Markov chain is said to be *homogeneous*. Conversely, if the statistical behavior of the source changes over time, we need to resort to a *non-homogeneous* Markovian chain, whose transition matrix varies with the time step n. Unless otherwise specified, in the following we assume homogeneous Markov chains.

Note that, in a standard discrete-time Markov model, state transitions are considered at discrete, equally spaced time instants. In some applications, however, transitions between operational states occur at random time instants, and the duration for which the source remains in a given state may itself be a random variable. In this case, the discrete-time Markov chain can be used to describe the sequence of states visited by the source, while a separate stochastic model describes the sojourn time in each state. Models of this type are referred to as *semi-Markov models*.

3.5.1 Markov Modulated Poisson Process

The Markov-Modulated Poisson Process (MMPP) is a class of Markovian models particularly relevant in the context of source modeling. Basically, the model represents traffic generation as events of a non-homogeneous Poisson process, whose rate changes in discrete time steps driven by the stochastic state transitions of the underlying Markov model.

Mathematically, each state $i \in \Omega$ of the Markov process is associated with a different traffic generation rate λ_i of the Poisson process that models the traffic generation events. The MMPP is then defined by the pair $(\mathbf{P}, \boldsymbol{\lambda})$, where $\mathbf{P}$ is the transition matrix of the Markov chain, while $\boldsymbol{\lambda}$ is the vector of rates assigned to each state of the Markov chain.

A common type of MMPP is the so-called Interrupter Poisson Process, which consists of a two-states Markov chain, where state 0 is associated with a zero generation rate $\lambda_0 = 0$, while state 1 is associated with a positive generation rate $\lambda_1 = \lambda > 0$. This model corresponds to an On-Off model, where state 0 is OFF and state 1 is ON. The sojourn times in each state are i.i.d. geometric r.v.s with means $1/P_{0,1}$ and $1/P_{1,0}$ for state OFF and ON, respectively. The generation patter in ON state, finally, is Poisson of rate λ. Therefore, the long-term average rate of such a source is given by

$$\mathrm{G} = \frac{\lambda T_{\mathrm{ON}}}{T_{\mathrm{ON}} + T_{\mathrm{OFF}}} = \frac{\lambda P_{0,1}}{P_{1,0} + P_{0,1}} = \lambda p_{\mathrm{ON}}$$

where

$$p_{\mathrm{ON}} = \frac{P_{0,1}}{P_{1,0} + P_{0,1}}$$

is the steady-state probability of being in the ON state.

More generally, for any MMPP $(\mathbf{P}, \boldsymbol{\lambda})$ the long-term traffic is given by

$$\mathrm{G} = \sum_{i \in \Omega} \lambda_i p_i$$

where $\mathbf{p} = [p_i]$ is the steady-state probability distribution of the Markov process and, in turn, of the Poisson process rate.

3.6 Examples of source models

3.6.1 Voice and Audio signals

The simplest form to store digital audio is uncompressed Pulse Code Modulation (PCM). In this case, the audio signal is sampled with a certain frequency B, and each sample is quantized on $L = 2^b$ levels, and represented by b bits. Since human voice can reach frequencies of up to 4 kHz, a sampling frequency of $B = 8$ kHz and $L = 256$ quantization levels ($b = 8$ bits per sample) are sufficient to guarantee an excellent quality. In fact, digital phone calls originally used PCM coding with these parameters, generating a continuous stream of bits (CBR) with a bitrate of $R = 64$ kbit/s. Successively, more efficient coding techniques have been introduced, reducing the required bitrate to few kbit/s, though at the cost of a perceivable degradation of the voice quality. Some codecs also perform *silence suppression*. In this case, when the power of the audio signal drops below a certain threshold (slightly larger than the background noise power), the codec generates control message to notify the receiver that the sender is silent, and that no more packets will be sent until the source will start generating new audio content. Upon receiving this information, the receiver may keep reproducing a noise signal, called *comfort noise*, to maintain the same level of background noise that is perceived during active conversation. This trick is used to give to the users the perception that the communication is still ongoing, since the sudden interruption of any sound may be wrongly interpreted as an abrupt termination of the conversation.

For general audio signals, the sampling rate must be at least 40 kHz, because humans can hear sounds with frequencies up to 20 kHz. For digital music recoding, the sampling rates are slightly higher, namely 44.1 kHz for CD audio or 48 kHz for DVD. The quantization of the audio sample introduces a certain error, which is perceived as acoustic noise. The power of such a noise component is reduced by approximately 6 dB for any extra bit used in the sample coding. For example, CD audio uses 16 bits per sample, and therefore will have quantization noise approximately 96 dB below the maximum possible sound pressure level. However, the CD audio stream requires a much higher bitrate than standard voice calls, being equal to $R = 44,1 \times 16 \times 2 = 1,411$ Mbit/s (the factor 2 accounts for the stereo channels). Notice that, 70 minutes of audio signal (approximately, the total duration of the music tracks of a disk) corresponds to $S = R \times 70 \times 60/8 \simeq 740$ Mbytes, which was the typical data capacity of a CD when used as storage.

The bitrates can be greatly reduced using audio *codec*, which performs audio compression/decompression. Lossless codecs reduce the bitrate by a factor from 30% to 50% by eliminating some redundancy in the signal, but without discarding information (lossless compression). In this case, the original PCM stream can be recovered with no change in quality.

Much higher compression ratios can be obtained by means of lossy audio codec, such as MP3 or AAC. In this case, the codecs first get rid of audio components that are hardly perceivable by human hearing, using principles from psychoacoustics. Then, lossless compression can be applied to the remainder of the audio signal to further reduce the bitrate. Lossy audio compression can achieve 75-95% reduction in file size/stream bitrate, at the cost of a lower audio quality, in particular in case of high compression ratios.

3.6.2 Internet Traffic Models

Empirical studies have shown that Internet traffic exhibits characteristics that challenge traditional Poisson models, such as:

- *Long-Range Dependence*: Autocorrelation decays slowly following a power-law.
- *Self-Similarity*: Traffic retains similar statistical properties across different time scales.
- *Heavy-Tailed Distributions*: Values much larger than the median occur with significant probability.

This self-similar and long-rang dependent behavior is often observed in aggregated network traffic, such as traffic generated by web requests or streaming services.

Advanced models incorporating fractional Brownian motion or heavy-tailed distributions are necessary to capture these features accurately. Hybrid models combine features of multiple mathematical models to better capture the diverse and multi-scale nature of Internet traffic. For example, an on-off model can be combined with a Markovian framework to capture both burstiness and correlation, or a renewal process can be augmented with self-similar properties to reflect long-range dependence. Common models include the following.

3.6.2.1 Fractional Gaussian Noise Fractional Gaussian Noise (FGN) is a stochastic process frequently employed to model self-similar and long-range dependent Internet traffic. It is derived as the discrete-time increment process of Fractional Brownian Motion (FBM), a continuous-time Gaussian process defined by its stationary increments and variance scaling with the Hurst parameter $H \in (0, 1)$. Specifically, the increments of FBM are stationary and satisfy $\mathrm{E}\left[(B_H(t) - B_H(s))^2\right] = \sigma^2|t-s|^{2H}$, where $B_H(0) = 0$. The discrete version, FGN, is defined as $X_t = B_H(t+1) - B_H(t)$ for integer t, where X_t represents the traffic increment over a unit time step.

A central property of FGN is the role of the Hurst parameter H, which determines the nature of correlations in the process. When $H = 0.5$, FGN reduces to standard Gaussian white noise, exhibiting no long-range dependence or self-similarity. For $H > 0.5$, the process exhibits long-range dependence, where correlations decay hyperbolically with lag, reflecting the burstiness and persistence typical of Internet traffic. In contrast, $H < 0.5$ corresponds to anti-persistent behavior, which is less common in Internet traffic contexts. The power spectral density of FGN also highlights its self-similar nature, following a power-law decay with $S(f) \sim |f|^{1-2H}$ as $f \to 0$, indicating the dominance of low-frequency components.

Despite its strengths, FGN has limitations. Its assumption of Gaussian increments may not fully capture the heavy-tailed behavior of packet inter-arrival times or file sizes often observed in real Internet traffic. Additionally, while it is well-suited for modeling aggregated traffic, it may not accurately represent finer-grained details of individual flows. Calibrating the Hurst parameter H and variance σ from empirical data can also be challenging, particularly for traffic with highly variable characteristics.

Finally, we observe that a point process generated using FGN for its inter-arrival times is not a renewal process because the inter-arrival times are correlated (except in the trivial case where $H = 0.5$).

3.6.2.2 Heavy-Tailed Renewal Processes Heavy-tailed renewal processes are a class of stochastic processes that extend the classical renewal process by allowing inter-arrival

times to follow heavy-tailed probability distributions. These processes are particularly useful for modeling Internet traffic, where inter-arrival times and event durations often exhibit significant variability and heavy-tailed characteristics.

In a heavy-tailed renewal process, therefore, the distribution of the inter-arrival times A_i is characterized by a tail that decays slower than exponentially, typically following a power-law behavior. Mathematically, this is expressed as:

$$\mathrm{P}\left[A_i > t\right] \sim c \cdot t^{-\alpha}, \quad t \to \infty, \quad 0 < \alpha < 2,$$

where $c > 0$ is a constant and α is the tail index. The smaller the value of α, the *heavier* the tail, implying higher variability and a greater probability of very large inter-arrival times.

Heavy-tailed renewal processes have several unique properties that make them suitable for Internet traffic modeling. For $1 < \alpha < 2$, the inter-arrival times have finite mean but infinite variance, capturing the burstiness of Internet traffic. For $\alpha \leq 1$, even the mean becomes infinite, reflecting extreme variability.

When aggregated over multiple flows or time scales, heavy-tailed renewal processes can exhibit self-similarity and long-range dependence, two properties commonly observed in real-world traffic traces. Finally, due to the high variability in inter-arrival times, events tend to cluster, creating periods of high activity interspersed with long inactive periods. This reflects the bursty nature of Internet traffic, such as traffic generated by file transfers, video streaming, or web browsing.

Remark In general, the superposition of independent renewal processes is not itself a renewal process, since successive inter-arrival times of the superposed process need not be independent. A notable exception is the superposition of independent Poisson processes, which is again Poisson.

3.6.2.3 Fractional ARIMA Functional ARIMA (FARIMA) models are an extension of the traditional Auto-Regressive Integrated Moving Average (ARIMA) processes, specifically designed to incorporate long-range dependence and self-similarity.

A FARIMA(p, d, q) process is defined as:

$$S_t - \sum_{k=1}^{p} \varphi_k S_{t-k} = \epsilon_t + \sum_{h=1}^{q} \theta_h \epsilon_{t-h} \tag{3.7}$$

where

$$S_t = \sum_{k=0}^{\infty} \frac{\Gamma(d+1)}{\Gamma(k+1)\Gamma(d-k+1)} (-1)^k X_{t-k}\,, \tag{3.8}$$

and $\Gamma(\cdot)$ indicates the Generalized Gamma function. In (3.7), the parameters φ_i and θ_i are those of the autoregressive (AR) and moving average (MA) components, respectively, while ϵ_t is white noise with zero mean and variance σ^2. S_t in (3.8) is a weighted sum of the current and past values of the time series, where the weights decay according to the fractional differencing parameter d. Specifically, for $0 < d < 1$, the weights decay more slowly than in the integer differencing case, introducing long-term memory into the process. This means that distant past values of the series still influence the current value, capturing long-range dependence. The closer d is to 0, the weaker the memory and the closer the process is to white noise. The closer d is to 1, the stronger the memory and the closer the process is to a unit root non-stationary process. Therefore, the fractional

differencing terms S_t allow the process to capture both short-term correlations (via the AR and MA terms) and long-term dependencies (via the fractional d term). This makes FARIMA particularly suitable for modeling traffic patterns that exhibit both burstiness over short time scales and persistence over longer durations.

To be noticed that FARIMA models assume stationarity and may not capture non-stationary traffic trends or abrupt changes in traffic patterns due to events such as flash crowds or sudden shifts in user behavior. Additionally, the calibration of FARIMA parameters, especially the fractional differencing parameter d, can be computationally intensive and requires careful statistical fitting to match observed traffic data.

3.6.3 Machine-Type Communications

MTC represent a key component of modern communication networks, especially in the context of IoT applications. Unlike human-centric traffic, MTC traffic is characterized by sporadic activity, short payloads, and a massive number of devices, often exhibiting unique patterns such as synchronized transmissions and correlated events.

3.6.3.1 Periodic and/or sporadic MTC traffic. Sporadic traffic can be modeled as a PP with:

$$\mathcal{T} = \{\tau_i\}, \quad \tau_i \in \mathbb{R}^+, \quad i = 1, 2, \ldots$$

where τ_i represents the time instants when a device generates a packet. A simple model assumes Poisson arrival, where inter-arrival times $\mathcal{A} = \{A_i\}$ are independent identically distributed (i.i.d.) exponential r.v.s with mean λ^{-1}, representing the average packet generation rate of the device.

However, in many MTC applications, devices exhibit periodic traffic patterns, which can be captured using a deterministic inter-arrival time model:

$$\mathcal{A} = \{A_i = T\}, \quad \forall i\,,$$

where T is the fixed inter-arrival time between consecutive transmissions.

3.6.3.2 Bursty MTC traffic. Certain MTC applications, such as sensor networks, generate bursty traffic due to events like environmental anomalies. A bursty MTC source can be modeled as a MWPP:

$$(\mathcal{T}, \mathcal{B}, \mathcal{W}),$$

where B_i represents the number of packets (or data units) generated in burst i, and $W_{i,k}$ the packet sizes. For example, the burst size B_i can follow a geometric distribution:

$$\mathrm{P}\left[B_i = k\right] = p(1-p)^{k-1}, \quad k \geq 1\,,$$

where p is the probability of generating exactly one packet during a burst. The inter-arrival times $\mathcal{A}$ between bursts can be modeled as exponentially distributed, reflecting the random nature of event-driven bursts.

3.6.3.3 Massive Access with Correlated Events. In scenarios with a massive number of MTC devices (e.g., smart meters or IoT sensors), events such as power grid failures or natural disasters can trigger synchronized transmissions from many devices. This phenomenon, often referred to as *correlated access*, can be modeled using a *superposed point process*:

$$\mathcal{T}_{\text{total}} = \bigcup_{j=1}^{N} \mathcal{T}_j,$$

where $\mathcal{T}_j$ represents the point process of the jth device, and N is the total number of devices.

Each individual device may follow a Poisson process with rate λ_j, and their superposition results in a total arrival rate:

$$\Lambda_{\text{total}} = \sum_{j=1}^{N} \lambda_j.$$

In addition, the traffic during correlated events can be modeled using a *conditional intensity function*, where the arrival rate increases significantly upon the detection of an event:

$$\lambda(t) = \lambda_0 + \lambda_{\text{event}} \cdot \chi \left\{ t \in \mathcal{T}_{\text{event}} \right\},$$

where $\mathcal{T}_{\text{event}}$ represents the time interval of the event, λ_0 is the baseline arrival rate, and λ_{event} is the increased rate during the event.

3.6.3.4 Packet-Level Traffic with Priorities

In MTC scenarios, devices often generate traffic with different priorities, such as critical alarms or periodic updates. These priorities can be incorporated into the Marked and Weighted Point Process (MWPP):

$$(\mathcal{T}, \mathcal{B}, \mathcal{W}),$$

where $W_{i,k}$ represents the priority weight of the kth packet in the ith arrival. For example, higher-priority traffic may have larger weights:

$$W_{i,k} = \begin{cases} w_{\text{high}}, & \text{if high priority,} \\ w_{\text{low}}, & \text{otherwise.} \end{cases}$$